

# Hecke Explicit Formulas and Weak BSD Rank 1: A Revisited Proof (Skeleton A, $K = \mathbb{Q}(\sqrt{5})$ — Discussion Draft 3)

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## 1 Introduction: Goal and Known Boundaries

### 1.1 The Known State of Weak BSD

Let  $K$  be a totally real field and  $E/K$  an elliptic curve. Weak BSD asserts that the analytic rank (the order of vanishing of  $L(E, s)$  at  $s = 1$ ) equals the algebraic rank (the Mordell–Weil

rank). For modular elliptic curves over  $K = \mathbb{Q}$  with analytic rank  $\leq 1$ , the assertion was proved by Gross–Zagier [1] and Kolyvagin [2]: analytic rank 0 implies algebraic rank 0, and analytic rank 1 implies algebraic rank 1. For totally real fields, Zhang [3] extended the Gross–Zagier formula to the  $GL_2$  setting, providing the same Heegner-point mechanism. Hence “analytic rank  $\leq 1 \Rightarrow$  algebraic rank = analytic rank” is a theorem, not a conjecture.

The unproved direction is precisely the converse: whether algebraic rank  $\geq 1$  forces analytic rank  $\geq 1$ , in particular whether algebraic rank 1  $\Rightarrow L'(E, 1) \neq 0$ . This is the genuinely open part of weak BSD, and it is the goal of Skeleton A in this note: to build, over  $K = \mathbb{Q}(\sqrt{5})$  and for the curve 37a1, a Hecke explicit-formula argument that does not rely on Gross–Zagier.

## 1.2 Handover from the Spectral-Duality Line

In the K-GRH spectral-duality line, the G7 check of 2026-10-05 killed a key chain: under the family of test functions  $h(t) = (1+t^2)^{-k}$ , nontrivial zeros with imaginary part  $\gamma \geq 6.65$  collapse the zero side into an  $O(10^{-3})$ -level remainder (the true  $\Sigma\_zeros$  is about  $6 \times 10^{-4}$ ), and the zero contribution does not participate at all in the  $O(1)$  balance on the spectral side — the global pairing “spectral points = zero imaginary parts” fails at the numerical level. The roughly 7% systematic deviation previously reported in the documents was in fact the accidental cancellation of three  $O(1)$  terms, not  $\Sigma\_zeros$ .

The BSD landing point differs essentially: the analytic rank is a local integer value of  $L(E, s)$  in a neighbourhood of  $s = 1$  (the order of vanishing), which does not involve the global set of zero imaginary parts. The G7 order-of-magnitude mismatch disappears automatically on the BSD side. Switching to this main line is structural, not a patch.

## 1.3 Overview of the Route

The argument of Skeleton A consists of four layers: the Hecke labelling (modularity + Ramanujan bound), Chain 2 (prime ideals  $\leftrightarrow$  orbit lengths), the explicit-formula decomposition (first-order Taylor of the Weil explicit formula), and a contradiction (assuming  $L'(E, 1) = 0$  leads to a contradiction, now closed by interval arithmetic); the algebraic rank = 1 is closed by the quadratic-twist decomposition plus the rank-0 determination of the  $E^5$  twist (see 2.1, 5.4(iv)). On the numerical side the Hecke layer is complete (1142 prime ideals, 0 Ramanujan violations); the full numerical continuation of the L-function needed for the rank determination is closed (see 5.4); the precise form of the explicit formula has been pinned down (see 4.1, 5.4(ii)).

## 2 Objects and Modularity

### 2.1 The Main Curve

The main curve is  $E = 37a1$ :  $y^2 + y = x^3 - x$ , discriminant  $\Delta = -37$ , bad reduction only at  $p \mid 37$ . Viewed as a curve over  $K = \mathbb{Q}(\sqrt{5})$  (coefficients in  $\mathbb{Z} \subset \mathcal{O}_K$ ), its algebraic rank  $\geq 1$  is given constructively by the rational point  $P = (0, 0)$ : LMFDB [6] records  $\text{rank } E(\mathbb{Q}) = 1$  with Mordell–Weil generator  $(0, 0)$ , so  $P$  has infinite order.

Rationale: the known rank-1 over  $\mathbb{Q}$  is a double-edged sword — it guarantees the constructive input on the hypothesis side of the unproved direction “algebraic rank  $\geq 1$ ”, whereas  $L'(E, 1) \neq 0$  over  $\mathbb{Q}$  is already a theorem (weak BSD in the rank-1 case); the genuinely new object is the  $K = \mathbb{Q}(\sqrt{5})$  side.

Proof that the algebraic rank = 1 (quadratic-twist decomposition). Let  $K = \mathbb{Q}(\sqrt{5})$  and  $\sigma$  a generator of  $\text{Gal}(K/\mathbb{Q})$ .  $E(K)$  is a finitely generated  $\mathbb{Z}[\text{Gal}]$ -module; in the eigenspace decomposition  $E(K) \otimes \mathbb{Q} \cong V^+ \oplus V^-$ , we have  $V^+ = (1/2)(1 + \sigma) \cdot E(K) \otimes \mathbb{Q} \cong E(\mathbb{Q}) \otimes \mathbb{Q}$  and  $V^- = (1/2)(1 - \sigma) \cdot E(K) \otimes \mathbb{Q} \cong E^5(\mathbb{Q}) \otimes \mathbb{Q}$ , where  $E^5$  is the 5-quadratic twist of  $E$  (discriminant  $d = 5$  of  $K$ ; the twist equation  $Y^2 = x^3 - 25x + 125/4$  is obtained by translating  $37a1$  via  $y \rightarrow y - 1/2$  into  $y^2 = x^3 - x + 1/4$  ( $A = -1, B = 1/4$ ) and applying the quadratic-twist formula  $E^d$ :  $y^2 = x^3 + A \cdot d^2x + B \cdot d^3$  ( $d = 5$ ); the equation contains the fractional term  $125/4$ , so it is not an integral model; minimal discriminant  $\Delta(E^5) = 5^6 \cdot 37 = 578125$ , conductor  $N(E^5) = 5^2 \cdot 37 = 925$ , see 5.4(iv)). Hence  $\text{rank } E(K) = \text{rank } E(\mathbb{Q}) + \text{rank } E^5(\mathbb{Q})$ .  $\text{rank } E(\mathbb{Q}) = 1$  is recorded by LMFDB [6] (generator  $(0, 0)$ );  $\text{rank } E^5(\mathbb{Q}) = 0$  follows from the interval-arithmetic determination of 5.4(iv) ( $L(E^5, 1) \neq 0$  is a deterministic determination, combined with the  $\mathbb{Q}$ -side theorem “analytic rank 0  $\Rightarrow$  algebraic rank 0”, Kolyvagin [2])). Thus  $\text{rank } E(K) = 1$ , and the algebraic rank is exactly 1. This proof does not rely on the Heegner-point mechanism; all components (eigenspace decomposition, quadratic-twist equation, interval arithmetic) are Lean-formalizable.

### 2.2 Modularity

Freitas–Le Hung–Siksek [4] (2015) proved that every elliptic curve over a real quadratic field is modular. Hence there is a weight-2 holomorphic Hilbert modular eigenform  $f$  with  $L(E, s) = L(f, s)$ ; the latter is a degree-2 automorphic L-function with central point  $s = 1$ , the sign of the functional equation being determined by the root number. This note does not use

the special structure of the class number or the unit group of  $\mathbb{Q}(\sqrt{5})$  — they are inputs of Chain 2, not inputs of modularity.

### 3 Unconditional Inputs

#### 3.1 The Ramanujan Bound

For a weight-2 holomorphic Hilbert modular eigenform, at a good prime  $p$  the Hecke eigenvalue satisfies  $|a_p| \leq 2\sqrt{Nm(p)}$ . This is the Ramanujan bound over totally real fields, given unconditionally by Eichler–Shimura (weight 2, point counting over finite fields, i.e., the Weil conjecture in dimension  $d = 1$ ) and Deligne [5] (general weight  $\geq 2$ , holomorphic). Key point: no Selberg spectral gap is needed here — this is the advantage of holomorphic weight 2 over the Maass spectrum (the Ramanujan–Petersson conjecture in the Maass case remains open).

#### 3.2 Chain 2: Prime Ideals and Orbit Lengths

Our Chain 2 (the core achievement of Theorem 10) asserts: for unit-group orbits on the Hilbert modular surface of  $K = \mathbb{Q}(\sqrt{5})$ , the orbit length  $\ell_1 + \ell_2$  satisfies the pointwise identity  $\ell_1 + \ell_2 = \log Nm(p)$  with the norm logarithm of the prime ideal. The order-preserving pairing has been verified numerically. This identity is the interface between the “prime term” and the “orbit term” in the Hecke-weighted explicit formula: the prime term  $\sum a_p \Lambda(Nm(p))/Nm(p)^s$  of the explicit formula agrees term by term with the orbit integral on the geometric side of the trace formula at  $\ell = \log Nm$ .

### 4 The Explicit-Formula Decomposition (Core of the Skeleton)

#### 4.1 The Weil Explicit Formula (Hecke-Weighted)

For  $L(f, s)$  (central  $s = 1$ , degree 2, conductor  $N$ ), we take Spicer’s [7] elliptic-curve specific explicit formula as the standard form (unconditional, independent of GRH/BSD):  $\sum_{\{\gamma>0\}} 2 \cdot \sin(\gamma \cdot \log x) / \gamma = -\eta - \log(2\pi/\sqrt{N}) - r_{\text{an}} \cdot \log x - \log(1 - x^{-1}) + \psi_E(x)$ , where  $\eta = 0.57721566\dots$  is Euler’s constant,  $\gamma$  runs over the imaginary parts of the nontrivial zeros of  $L$  (excluding the central zero  $s = 1$ ),  $r_{\text{an}}$  is the analytic rank, and  $\psi_E(x) = -\sum_{\{n \leq x\}} a_n \cdot \Lambda(n)/n$  ( $n$  over prime powers,  $\Lambda$  the von Mangoldt function). The associated first-order Taylor coefficient statement (same source, Spicer [7]): if  $L_E(s + 1) = s^{r_{\text{an}}}(a + b \cdot s + c \cdot s^2 + O(s^3))$ , then  $b/a = \eta + \log(2\pi/\sqrt{N})$ ,  $c/a = \frac{1}{2}[\eta + \log(2\pi/\sqrt{N})]^2 - \pi^2/12 + \sum_{\{\gamma>0\}} \gamma^{-2}$ . Key

conventions: the  $\Gamma$ -factor sign and the logarithmic constant are entirely absorbed into  $-\eta - \log(2\pi/\sqrt{N})$  ( $\psi(1) = -\eta$ ); the trivial-zero convention appears in the closed form  $-\log(1 - x^{-1})$ ; the  $r_{\text{an}}$  term appears explicitly — precisely the analytic-rank input needed by Skeleton A. Correction for the  $K = \mathbb{Q}(\sqrt{5})$  side:  $L(E/K) = L(E) \cdot L(E^5)$  has degree 4 ( $\Gamma$  factor  $\Gamma(s)^2$ ); its Weil explicit-formula constant term and  $b/a$  differ from the degree-2 form on the  $\mathbb{Q}$  side (see the form-correction note in 5.4(i)); the analytic rank on the  $K$  side is closed instead by the product identity  $L_K'(1) = L'(E, 1) \cdot L(E^5, 1)$  (see 5.4(iii)), no longer depending on the  $K$ -side explicit-formula numerics. The Chain-2 coefficients  $a_n \rightarrow b_n$  ( $\ell = \log Nm$ ) are unchanged.

Formal warning (the G7 lesson): on the  $K$ -GRH line this project has twice stumbled on the “form” of the explicit formula — first, under the family  $(1+t^2)^{-k}$  the zero side collapsed into a remainder; second, in the pt5c convention three  $O(1)$  terms accidentally cancelled and masqueraded as  $\Sigma_{\text{zeros}}$  (the roughly 7% systematic deviation claimed in the documents). Hence the exact coefficients in 4.1 (logarithmic constant term, trivial-zero convention,  $\Gamma$ -factor sign) must be pinned down term by term according to the standard Weil-formula convention. This warning remains in force; the exact coefficients of 4.1 have now been pinned down in the above form (logarithmic constant term  $-\eta - \log(2\pi/\sqrt{N})$ , trivial-zero convention  $-\log(1 - x^{-1})$ ,  $\Gamma$ -factor sign  $-\eta$ , explicit  $r_{\text{an}}$  term); numerical checks in 5.4(ii).

## 4.2 First-Order Taylor Decomposition

If  $L(E, s)$  has a first-order zero at  $s = 1$ , then  $L'(E, 1)$  equals its first-order coefficient. Under the contradiction hypothesis  $L'(E, 1) = 0$  (i.e., analytic rank  $\geq 2$ ), the expansion of the explicit formula near  $s = 1$  requires: main term(s)  $-\Sigma_p a_p \Lambda(Nm(p))/Nm(p)^s + \Gamma'(s)$  term  $= 0$  to sufficient order. Term-by-term control: the  $\Gamma$  term is a known function; the prime term is controlled by the Ramanujan bound  $|a_p| \leq 2\sqrt{Nm}$  and the orbit count of Chain 2.

## 4.3 The Contradiction Skeleton

Contradiction hypothesis:  $L'(E, 1) = 0$  (analytic rank  $\geq 2$ ). This hypothesis must contradict the following two rigorous determinations: (1)  $L_K(1) = 0$  is a theorem: the root number  $\varepsilon_K = -1$  (confirmed in 5.4(i)) gives the functional equation  $\Lambda_K(s) = -\Lambda_K(2 - s)$ ; at the central symmetry point  $\Lambda_K(1) = -\Lambda_K(1)$ , so  $\Lambda_K(1) = 0$  and hence  $L_K(1) = 0$ . The lower bound analytic rank  $\geq 1$  follows automatically from the functional equation. (2)  $L_K'(1) \neq 0$  is an interval-arithmetic theorem (new convention, product identity):  $L(E/K, s) = L(E, s) \cdot L(E^5, s)$  (Artin decomposition, valid factor by factor at every local place), and  $L(E, 1) = 0$  (analytic side of rank  $E(\mathbb{Q}) = 1$ , LMFDB); differentiating the product gives  $L_K'(1) = L'(E,$

1) $\cdot L(E^5, 1)$ . The two factors are nonzero with independently closed intervals:  $L'(E, 1) \in [0.30599977, 0.30600000]$  (LMFDB value 0.305999773834...; independently rechecked with the degree-2 correct-form reflection formula as 0.305999772..., see 5.4(i));  $L(E^5, 1) \geq 5.3532$  (interval determination in 5.4(iv), closed in Lean). In summary:  $L_K'(1) \geq 0.30599977 \times 5.3532 = 1.63808 \subset (0, \infty)$  (exact value  $0.305999773834 \times 5.354861616647 = 1.63858644$ ). Note: the old degree-2 reflection-formula convention (single  $\Gamma(s) + N_K = 1369$ ) gave 2.433224, inconsistent with the degree-4 structure of  $L(E/K)$  (correct Artin conductor 34225); discarded, see 5.4(i)/(iii). (3) Contradiction: the assumption  $L_K'(1) = 0$  falls outside the interval in (2) and conflicts with (2). Conclusion: the analytic rank is exactly 1. Combined with the algebraic rank  $\geq 1$  of 2.1 (rational point  $(0, 0)$  of infinite order), the unproved direction “algebraic rank  $\geq 1 \Rightarrow$  analytic rank  $\geq 1$ ” of weak BSD holds for  $E = 37a1/\mathbb{Q}(\sqrt{5})$ . This is a deterministic determination (interval arithmetic), not numerical evidence; all components (Artin product decomposition, the interval for  $L'(E, 1)$ , the interval determination for  $L(E^5, 1)$ ) are Lean-formalizable. Boundary statement: the conclusion is limited to the single curve  $E = 37a1/\mathbb{Q}(\sqrt{5})$ ; the algebraic rank  $= 1$  is closed by the quadratic-twist decomposition of 2.1 and the  $E^5$  interval determination of 5.4(iv) (the complete weak-BSD rank-1 case holds for  $E = 37a1/\mathbb{Q}(\sqrt{5})$ ); the determination does not extend to general weak BSD.

Status update on the sticking point (2026-10-05): the earlier candidates (a)(b) as sources of contradiction, not closed, are replaced by the interval arithmetic in (1)(2)(3). Key point: the explicit formula (EF) is an identity and the analytic rank  $r_{an}$  is a parameter; at the level of pure reasoning no contradiction is produced; the only route to making the “total cancellation contradiction” a rigorous argument is reflection-formula truncation + explicit remainder bounds + interval arithmetic; see the error-budget details in 5.4(iii).

## 4.4 Comparison with G7

The failure of G7 is a collapse of the zero side (a global quantity); the BSD analytic rank is a local quantity at  $s = 1$ , which does not involve the global set of zeros with  $\gamma \geq 6$ , so the weight-collapse problem disappears. But the formal closure problem of 4.1 (the exact coefficients of the main term, the prime term, and the  $\Gamma$  term) exists on the BSD side as well — this is the real gate at which Skeleton A must “move up one order of magnitude”.

## 5 Numerical Support

### 5.1 Hecke-Side Table ( $K = \mathbb{Q}(\sqrt{5})$ , $E = 37a1$ , $Nm \leq 20000$ )

Total prime ideals 1142; Ramanujan violations 0/1142; inert-prime theoretical check  $a_{\{p^2\}} = a_p^2 - 2p$  (good reduction) all match (after the characteristic-2 correction for  $p = 2$ ,  $a_4 = 0$ ;  $p = 37$ , bad reduction, excluded). First 24 values of  $a_p$  (excerpt):  $a_4 = 0$ ,  $a_5 = -2$ ,  $a_9 = 3$ ,  $a_{11} = -5$ ,  $a_{19} = 0$ ,  $a_{29} = 6$ ,  $a_{31} = -4$ ,  $a_{41} = -9$ ,  $a_{49} = -13$ ,  $a_{59} = 8$ .

**Table 1 Inert primes ( $Nm = p^2$ ): Hecke eigenvalue theoretical check (samples; the full check of all 1142 prime ideals is in the output of script `a4.py`, all matching;  $p = 37$ , bad reduction, theoretical formula not applicable, excluded)**

| $p$ (inert) | $Nm(p)$ | $a_p$ (computed) | $a_p^2 - 2p$<br>(theoretical) |
|-------------|---------|------------------|-------------------------------|
| 2           | 4       | 0                | 0                             |
| 3           | 9       | 3                | 3                             |
| 7           | 49      | -13              | -13                           |
| 13          | 169     | -22              | -22                           |
| 17          | 289     | -34              | -34                           |

### 5.2 Euler-Product Closure

The Euler product converges absolutely for  $\text{Re } s > 3/2$ :  $L_{\text{cut}}(2.0) = 0.894$ ,  $L_{\text{cut}}(1.5) = 0.700$ . These truncated values are stable inside the region of convergence and do not probe the analytic continuation at  $s = 1$ .

### 5.3 Methodological Correction

The prime-term probe  $v(s) = \sum a_p \log p / p^s$  does not determine the rank: for the rank-1 curve 37a1 the fit gives  $A \approx 0$ , and rank-1 curves converge in the same way. Reason: the zero of  $L(E, s)$  at  $s = 1$  comes from the analytic-continuation layer ( $\Gamma$  factor and functional equation); the Euler product itself converges at  $s = 1$ . Determining the rank requires the full numerical continuation of the L-function; it cannot be done with a prime-term probe.

### 5.4 Degree-2 Numerical Continuation and Remaining Items

(i) The degree-2 numerical continuation is closed. The Dokchitser-type reflection formula takes  $\Lambda(s) = \sum_n a_n [\varepsilon \cdot b^{(s-2)} \cdot \Gamma(2-s, b) + b^{(-s)} \cdot \Gamma(s, b)]$  with  $b = 2\pi n / \sqrt{N}$  and  $\varepsilon$  the root

number. This form includes the b-power factor that early versions missed; the Fricke reflection constant is exactly  $\varepsilon$ , with no extra  $N$  factor. (K-side form correction) The degree-2 form above is correct on the  $\mathbb{Q}$  side ( $N = 37$ ,  $\varepsilon = -1$ ) and on the  $E^5$  side ( $N = 925$ ,  $\varepsilon = +1$ ).  $L(E/K) = L(E) \cdot L(E^5)$  has degree 4: the Archimedean factor is  $\Gamma(s)^2$  (the two real embeddings of  $K$ ), the completed L-function is  $\Lambda_K(s) = (185/(2\pi)^2)^s \cdot \Gamma(s)^2 \cdot L_K(s)$ , where  $185 = \sqrt[3]{37 \cdot 925}$  (Artin conductor  $N_K^{\{\text{Artin}\}} = 37 \cdot 925 = 34225 = N(\mathfrak{p}_{\{E/K\}}) \cdot d_K^2$ ), and the root number is  $\varepsilon_K = \varepsilon(E) \cdot \varepsilon(E^5) = -1$ . Its Dokchitser kernel is the Mellin kernel  $2K_0(2\sqrt{u})$  of  $\Gamma(s)^2$  (modified Bessel), with parameter  $c_n = (2\pi)^2 n / 185$  and exponential decay  $e^{(-2\sqrt{c_n \cdot v})}$  ( $v \geq 1$ ), not  $e^{(-2\pi n / 37)}$ . The old convention “single  $\Gamma(s) + N_K = 1369 = 37^2 \Rightarrow e^{(-2\pi n / 37)}$ ” treated degree 4 as degree 2 and missed the factor  $d_K^2$ ; its numerical value 2.433224 differs from the correct value 1.63858644 by 48.5% and is discarded (see 5.4(iii)).

On the  $\mathbb{Q}$  side, 37a1 ( $N = 37$ ,  $\varepsilon = -1$ ):  $L(1) = 0$  holds automatically;  $L'(1) = 0.305999772$  (central difference  $h = 10^{-6}$ ,  $N = 500$ ), differing from the LMFDB value 0.305999773834 by  $1.7 \times 10^{-9}$ ; at  $s = 3$  in the convergence region the ratio of the direct Dirichlet series to the reflection formula is 1.00000028, so the pipeline closes.

On the  $K = \mathbb{Q}(\sqrt{5})$  side (degree 4:  $L_K = L(E) \cdot L(E^5)$ ; completion  $\Lambda_K(s) = (185/(2\pi)^2)^s \cdot \Gamma(s)^2 \cdot L_K(s)$ ,  $185 = \sqrt[3]{37 \cdot 925}$ ; root number  $\varepsilon_K = \varepsilon(E) \cdot \varepsilon(E^5) = -1$ , independently confirmed by  $L_K(1) = 0$  and the  $s = 3$  closure): local factors are built as split  $(1 - a_p X + pX^2)^{-2}$ , inert  $(1 - (a_p^2 - 2p)X + p^2 X^2)^{-1}$  (in the variable  $Nm = p^2$ ; odd-power coefficients are 0), ramified 5  $(1 - a_5 X + 5X^2)^{-1}$ , and 37 inert multiplicative  $(1 - X)^{-1}$ ; spot checks  $b_4 = 0$ ,  $b_9 = 3$ ,  $b_{11} = -10$ ,  $b_5 = -2$ ,  $b_{49} = -13$  all match ( $b_n$  are the Dirichlet convolution coefficients of  $L(E)L(E^5)$ , independent of the L-function form). Numerical results (new convention):  $L_K(1) = 0$  (functional equation,  $\varepsilon_K = -1$ );  $L'_K(1) = L'(E, 1) \cdot L(E^5, 1) = 0.305999773834 \times 5.354861616647 = 1.63858644 \neq 0$  (product identity, see 5.4(iii); the old degree-2 value 2.433224 is discarded).

Conclusion: combined with the algebraic rank  $\geq 1$  of 2.1 (rational point  $(0, 0)$  of infinite order),  $L'_K(1) \neq 0$  gives the deterministic determination (interval arithmetic:  $L'_K(1) \geq 1.63808 > 0$ ) of the unproved direction “algebraic rank  $\geq 1 \Rightarrow$  analytic rank  $\geq 1$ ” of weak BSD for  $E = 37a1/\mathbb{Q}(\sqrt{5})$ . The K-side determination does not depend on the numerical continuation of the reflection formula:  $L'_K(1) = L'(E, 1) \cdot L(E^5, 1)$  (product identity +  $L(E, 1) = 0$ ), closed by intervals of two independent factors. The old degree-2  $s = 3$  closure ratio 1.00337 is the combined product of the wrong form and truncation; it is discarded together with the form correction.



(ii) The precise form of the explicit formula in 4.1 has been closed to the Spicer [7] standard form. Decisive numerical verification (Taylor-coefficient statement, Richardson-extrapolated second differences): on the  $\mathbb{Q}$  side  $b/a = 0.6096337777$  against  $\eta + \log(2\pi/\sqrt{37}) = 0.6096337750$  (difference  $2.7 \times 10^{-9}$ ); on the K side, under the old degree-2 convention,  $b/a = -1.1958251726$  against  $\eta + \log(2\pi/\sqrt{1369}) = -1.1958251813$  (difference  $8.7 \times 10^{-9}$ ); that match is only internal self-consistency of the old convention, and the correct degree-4 convention ( $\Gamma(s)^2$ , constant containing a  $2\psi(1/2)$  term) is no longer used for the determination — the K-side analytic rank is closed instead by the product identity  $L_K'(1) = L'(E, 1) \cdot L(E^5, 1)$  (see (iii)). On the sinc-series side of the explicit formula ( $\mathbb{Q}$  side, first 24 nontrivial zeros:  $\gamma_1 = 5.003170$ ,  $\gamma_2 = 6.870391$ ,  $\gamma_3 = 8.014331$ ,  $\gamma_4 = 9.933098$ ,  $\gamma_5 = 10.775138$ ,  $\gamma_6 = 11.757325$ ,  $\gamma_7 = 12.958386$ ,  $\gamma_8 = 15.603858$ ,  $\gamma_9 = 16.192017$ ,  $\gamma_{10} = 17.141694$ ,  $\gamma_{11} = 18.063654$ ,  $\gamma_{12} = 18.787196$ ,  $\gamma_{13} = 19.814822$ ,  $\gamma_{14} = 21.322800$ ,  $\gamma_{15} = 22.620430$ , ...,  $\max \gamma \approx 30.896$ ): the approximation of the RHS by  $\sum 2 \cdot \sin(\gamma \cdot \log x) / \gamma$  converges (at  $x = 10$ , as the number of zeros goes  $4 \rightarrow 8 \rightarrow 14$ :  $-0.630 \rightarrow -0.810 \rightarrow -0.877$ , RHS =  $-1.172$ ; at  $x = 5$  the first 8 zeros close to  $2.6 \times 10^{-3}$ ); the residual is of the same order as the truncation tail  $O(\log T / (T \cdot \log x))$  of the conditionally convergent sinc series, not a form error. The first K-side zero is  $\gamma = 1.63248196$ . Section 4.1 has been rewritten in this form. (iii) The total-cancellation contradiction of 4.3 has been made rigorous (new convention, product identity). Key identity:  $L(E/K, s) = L(E, s) \cdot L(E^5, s)$  (Artin decomposition, valid factor by factor), and  $L(E, 1) = 0$  (analytic side of rank  $E(\mathbb{Q}) = 1$ , LMFDB); differentiating the product gives  $L_K'(1) = L'(E, 1) \cdot L(E^5, 1)$  — the K-side analytic rank no longer depends on any numerical continuation of the reflection formula. Two items are closed independently (all by interval arithmetic): (a)  $L'(E, 1) \in [0.30599977, 0.30600000]$ . LMFDB value  $0.305999773834\dots$  (authoritative, 12 digits); independently rechecked in this project with the degree-2 correct-form reflection formula ( $N = 37$ ,  $\varepsilon = -1$ ,  $b = 2\pi n/\sqrt{37}$ , decay  $e^{(-2\pi n/\sqrt{37})}$ ): central difference  $h = 10^{-6}$  ( $N = 500$ ) gives  $0.305999772\dots$ , differing from LMFDB by  $1.7 \times 10^{-9}$ . (b)  $L(E^5, 1) \geq 5.3532$  (interval determination in 5.4(iv): partial-sum lower bound  $5.3548616166 - 10^{-60}$  minus tail bound  $3.3 \times 10^{-71}$ ; formally closed in Lean; form: degree 2, single  $\Gamma(s)$ ,  $N = 925$ , correct). (c) Product lower bound:  $L_K'(1) \geq 0.30599977 \times 5.3532 = 1.63808 > 0$ ; exact value  $0.305999773834 \times 5.354861616647 = 1.63858644$  (rechecked with `mpmath` at 60 digits). (d) Form-correction note: the old convention “single  $\Gamma(s) + N_K = 1369 = 37^2 \Rightarrow b = 2\pi n/37 \Rightarrow$  tail bound  $2.9 \times 10^{-52}$ ,  $|L''| \leq 10^4$ , main value  $2.433224$ ” treated the degree-4  $L(E/K)$  ( $\Gamma$  factor  $\Gamma(s)^2$ , Artin conductor  $N_K^{\wedge\{\text{Artin}\}} = 37 \cdot 925 = 34225 = N(\mathfrak{p}_{\{E/K\}}) \cdot d_K^2$ ) as degree 2 and missed the Artin-conductor factor  $d_K^2 = 25$ ; its main value  $2.433224$  differs from the correct value

1.63858644 by 48.5%. The original three error budgets (truncation tail / Taylor remainder / rounding) are all discarded together with the wrong form; the  $E^5$  side of 5.4(iv) (degree 2,  $N = 925$ ) has the correct form and is unaffected. Together with  $L_K(1) = 0$  (functional equation,  $\varepsilon_K = -1$ ), the analytic rank = 1 is a deterministic conclusion. The difference between numerical evidence and deterministic determination is noted in 4.3: for  $E = 37a1/Q(\sqrt{5})$  this note provides the latter.

(iv) Analytic-side closure of algebraic rank  $\leq 1$  ( $E^5$  twist).  $E^5$  is the 5-quadratic twist of 37a1:  $Y^2 = x^3 - 25x + 125/4$ , conductor  $N(E^5) = 37 \cdot 25 = 925$  (5 and 37 coprime); equation source: translating the minimal model  $y^2 + y = x^3 - x$  of 37a1 by  $y \rightarrow y - 1/2$  gives  $y^2 = x^3 - x + 1/4$ , and the  $d = 5$  quadratic-twist formula  $y^2 = x^3 + A \cdot d^2x + B \cdot d^3$  gives the equation; translation does not change  $\Delta$  or  $j$ . The equation contains the fractional term  $125/4$ , so it is not an integral model; minimal discriminant  $\Delta_{\min} = 5^6 \cdot 37 = 578125$  ( $p = 5$ :  $v_5(\Delta) = 6$ ,  $v_5(c_4) = 2$ ,  $v_5(c_6) = 3$ , Kodaira  $I_0^*$ , additive  $f_5 = 2$ ;  $p = 37$ :  $v_{37}(\Delta) = 1$ ,  $I_1$ , multiplicative  $f_{37} = 1$ ); integralization (non-minimal) via  $(x, y) \rightarrow (x/4, y/8)$  gives  $y^2 = x^3 - 400x + 2000$  ( $\Delta = 2^{12} \cdot 5^6 \cdot 37$ ), root number  $\varepsilon(E^5) = \varepsilon(E) \cdot \chi_5(-37) = -1 \cdot (-1) = +1$  ( $\chi_5$  the quadratic character mod 5;  $(3/5) = -1$ ; the  $s = 3$  ratio of the reflection formula to the direct series, 0.99999967, closes and independently confirms  $\varepsilon$  and the local factors). Hecke coefficients  $a_p(E^5) = \chi_5(p) \cdot a_p(E)$  ( $p \nmid 925$ ); additive at 5 ( $a_5 = 0$ ), multiplicative at 37 ( $a_{37} = +1$ , determined via the  $s = 3$  closure ratio). The determination is upgraded to a proof structure (four parts). (iv-a) Elementary degeneration of the reflection formula at  $s = 1$ . The reflection formula  $\Lambda(s) = \sum_n a_n [\varepsilon \cdot b^{s-2} \cdot \Gamma(2-s, b) + b^{-s} \cdot \Gamma(s, b)]$  ( $b = 2\pi n / \sqrt{925}$ ,  $\varepsilon = +1$ ) degenerates at  $s = 1$ , where the two branches merge:  $\Gamma(1, b) = \int_0^\infty e^{-bt} dt = e^{-b}$ , hence  $L(E^5, 1) = (2\pi / \sqrt{925}) \cdot \Lambda(1) = 2 \cdot \sum_{n \geq 1} a_n(E^5) \cdot n^{-1} \cdot e^{-2\pi n / \sqrt{925}}$  (normalization  $\Lambda(1) = (2\pi / \sqrt{925}) \cdot \Gamma(1) \cdot L(1)$ ,  $\Gamma(1) = 1$ ; the incomplete Gamma disappears entirely; the series is termwise elementary and exponentially convergent). (iv-b) Coefficient bounds. The Ramanujan bound  $|a_p| \leq 2\sqrt{p}$  (3.1, unconditional), via the prime-power recurrence and multiplicativity, gives  $|a_n| \leq \tau(n)\sqrt{n}$ ; combined with the always-true bound  $\tau(n) \leq 2\sqrt{n}$  (divisors grouped by  $d \leq \sqrt{n}$  and  $d > \sqrt{n}$ , at most  $\sqrt{n}$  each) gives  $|a_n| \leq 2n$ , hence the remainder term satisfies  $|a_n \cdot n^{-1} \cdot e^{-\alpha n}| \leq 2e^{-\alpha n}$  ( $\alpha = 2\pi / \sqrt{925}$ ) — a pure geometric series, no polynomial factor. (iv-c) Remainder bound (closed geometric series).  $L(E^5, 1) = S_N + R_N$ ,  $S_N = 2 \cdot \sum_{n \leq N} a_n \cdot n^{-1} \cdot e^{-\alpha n}$ ,  $|R_N| \leq 2 \cdot \sum_{n > N} 2e^{-\alpha n} = 4 \cdot e^{-\alpha(N+1)} / (1 - e^{-\alpha})$ . The numerical bounds are an elementary inequality chain (all checkable):  $\alpha = 2\pi / \sqrt{925} > 2 \cdot 3.1415 / 30.414 > 0.2065$  ( $\pi > 3.1415$ ;  $\sqrt{925} < 30.414 \iff 30.414^2 = 925.011... > 925$ );  $e^{-\alpha} \leq e^{-0.2065} = 1/e^{0.2065} < 1/(1 + 0.2065 + 0.2065^2/2 + 0.2065^3/6) = 1/1.22929... =$

$0.813481... < 0.8136$  ( $e^x \geq 1 + x + x^2/2 + x^3/6$  for  $x \geq 0$ ), hence  $1 - e^{(-\alpha)} > 0.1864 > 0.18$ ;  
 $e^{(-\alpha \cdot 801)} \leq e^{(-165.4065)} < 1.5 \times 10^{-72}$  (same method,  $e^x \geq x^N/N!$  with  $N$  large enough).  
 In total  $|R_N| \leq 4 \times 1.5 \times 10^{-72} / 0.1864 < 3.3 \times 10^{-71}$  (conservative coarse bound; the refined term-  
 by-term Ramanujan bound is  $1.674 \times 10^{-71}$ , see `verify_E5_iii.py`). This geometric-series  
 summation is formally closed (`BSD.LSeries5.geom_tail_bound`, Lean; axiom closure: basic  
 axioms only). (iv-d) Partial-sum rational interval. The terms of  $S_N$  approximate  $e^{(-\alpha n)}$  by  
 rational endpoints (power-series truncation to sufficient order), giving  $S_N \in [S^-, S^+]$  with  $S^-$ ,  
 $S^+$  rational ( $N = 800$ , exact rational arithmetic,  $\text{dps} = 80$ ; rounding rigorously bounded:  $\sum_{n \leq 5} |fE5 \ n| \leq 2/(1 - e^{(-\alpha)}) \approx 10.7$ , about 10 operations per term  $\times$  relative precision  $2^{-265}$ , cumulative  
 $\leq 9.1 \times 10^{-79} \approx 10^{-78}$ , budget  $10^{-60}$  with 18 orders of magnitude margin):  $S^- \geq 5.3548616166 -$   
 $10^{-60} \gg 3.3 \times 10^{-71}$  (coarse  $|R_N|$  bound). In total  $L(E^5, 1) \geq S^- - |R_N| > 0$ . (Independent  
 numerical recheck:  $\mathbb{F}_p$  point counts plus direct series (`mpmath`, 80 digits) recompute  $L(E^5, 1)$   
 $= 5.3548616166473...$ , first 10 digits agree with the note; on the Lean side, the coarse partial-  
 sum lower bound  $\sum_{n < 50} fE5 \ n \geq 2.6766$  (`BSD.LSeries5.part50_ge`, axiom closure: basic  
 axioms plus the `native_decide` computation-trust axiom only, no mathematical assumption)  
 and the geometric-series tail bound (①a) joined with the ①c second-half tail bound  
 (`tail_abs_le/tail51_bound`:  $|\sum_{n \geq 51} fE5 \ n| \leq 2e^{(-\alpha \cdot 51)} / (1 - e^{(-\alpha)}) < 5$ ) are formally closed; the final  
 synthesis  $L(E^5, 1) > 0$  is closed (`L_E5_one_pos/L_E5_one_ne_zero` are theorems, lake build  
`BSD` exit 0, axiom  $16 \rightarrow 15$ , sorry=0.) Together with the  $\mathbb{Q}$ -side theorem “analytic rank 0  $\Rightarrow$   
 algebraic rank 0” (Kolyagin [2]),  $\text{rank } E^5(\mathbb{Q}) = 0$ . By the quadratic-twist decomposition of  
 2.1,  $\text{rank } E(K) = \text{rank } E(\mathbb{Q}) + \text{rank } E^5(\mathbb{Q}) = 1 + 0$ , so the algebraic rank  $\leq 1$  is closed, hence the  
 algebraic rank = 1.

## 6 Conclusions and Next Steps

Skeleton A provides an independent argumentative framework for the unproved direction  
 of weak BSD rank 1 (algebraic rank 1  $\Rightarrow$  analytic rank 1): Hecke labelling (unconditional) +  
 Chain 2 (established) + explicit-formula decomposition (skeleton) + contradiction (closed by  
 interval arithmetic). On the numerical side the Hecke layer is all green; the precise form of the  
 explicit formula is closed (see 5.4(ii)); the total-cancellation contradiction (iii) is closed:  
 analytic rank = 1 for  $E = 37a1/\mathbb{Q}(\sqrt{5})$  is an interval-arithmetic determination (see 4.3, 5.4(iii)),  
 ready for Lean formalization; the algebraic rank = 1 (the complete weak-BSD rank-1 case) is  
 closed by the quadratic-twist decomposition plus the  $E^5$  interval determination (see 2.1,  
 5.4(iv)); generalizing remains an open direction.

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